

DP Computer Science: A 1.1.1 CPU Architecture

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Describe the functions of the **Arithmetic Logic Unit (ALU)** and **Control Unit (CU)**.
- Describe the functions of the five registers: **PC, IR, MAR, MDR, AC**.
- Describe the functions and directions of the three buses: **address, data, control**.
- Differentiate between **single-core, multi-core**, and **co-processors**.
- Explain how these components connect and interact to process data.

Key Vocabulary (Tier 3)

Term	Definition
Arithmetic Logic Unit (ALU)	Performs mathematical calculations and logical operations.
Control Unit (CU)	Decodes instructions and directs the execution of operations.
Register	Tiny, extremely fast storage locations built directly into the CPU.
Bus	A set of parallel wires carrying addresses, data, or control signals.
Decode	The process of translating an instruction into control signals.
Multi-core processor	A single chip containing multiple independent processing units (cores).

Pedagogical Focus

- **Disciplinary Literacy:** Students will practice formulating precise technical explanations using the **FFF** (Feature, Function, Flow) and **PEAL** (Point, Evidence, Analysis, Link) structures.
 - **Differentiation:** Pre-teaching Tier 3 vocabulary provides a structured path for all students to articulate complex interactions without lowering academic demand.
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2. Lesson Sequence (60–75 minutes)

Part 1: Do Now (Retrieval) — ~10 mins

Activity: Students complete a short written recall task defining the difference between **data** and **instructions**. This activates prior knowledge from introductory courses, establishing the baseline for understanding what the CPU actually processes.

Part 2: Main Sequence — ~35 mins

A. The Big Picture (Diagrammatic Representation)

- Use the slide deck to introduce the CPU boundary, ALU, CU, and the three buses.
- Students annotate a blank diagram on their worksheet, drawing directional arrows to establish the macro-schema of the processor.

B. Zooming In (Vocabulary Pre-teaching)

- Students read the A 1.1.1 section of the InThinking eBook.
- They update their diagrams by labeling the five specific registers (PC, IR, MAR, MDR, AC) and adding internal connecting lines.
- Using this updated diagram, they complete the "Interacts with" quadrant on their Frayer models for each register.

C. Zooming Out (Types of Processors)

- Explicitly teach the definitions and distinctions between single-core, multi-core, and co-processors using the slide deck, expanding on the single-core schema just built.
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Part 3: Deliberate Practice — ~15 mins

Activity: Students work independently to write an **FFF (Feature, Function, Flow)** paragraph describing the specific role of the Control Unit (CU).

- **Requirement:** They must use at least three pre-taught vocabulary terms and explicitly reference its interaction with another component.
- **Support:** Provide sentence stems for the "Function" and "Flow" sections.
- **Challenge:** Students research how a co-processor (e.g., GPU) alters the interaction of main CPU components.

Part 4: Exit Check — ~5 mins

Activity: Students submit a single written response to an exam-style prompt:

"Describe the difference in function between the instruction register (IR) and the memory data register (MDR)." [4 marks]

The teacher reviews these responses to ensure students are not conflating addresses, data, and instructions before moving on to the machine instruction cycle.

3. Assessment & Exemplar Responses

Part 3 Exemplar (FFF Paragraph)

- Feature:** The Control Unit (CU) is the central coordinating component of the CPU.
- Function:** It fetches instructions from memory and decodes them into specific control signals that the hardware can execute.
- Flow:** It transmits these signals via the control bus to dictate the actions of other elements, such as signalling a register to release data to the ALU for calculation.

Part 4 Exemplar & Mark Scheme

"The Instruction Register (IR) holds the specific instruction that has just been fetched from memory and is currently being decoded. In contrast, the Memory Data Register (MDR) acts as a temporary buffer holding data values that have been read from RAM or are waiting to be written back to RAM."

Marks	Criteria
[1]	States what the IR holds.

Marks	Criteria
[1]	Explains its role in the current cycle.
[1]	States what the MDR holds.
[1]	Explains its role in the read/write process.

4. Engagement Activities (Supplementary)

If time allows, these can replace or supplement the main sequence:

1. **CPU Teardown:** Show a physical CPU or high-quality 3D model to visualise components.
2. **Role-Playing:** Groups act out a simplified fetch-decode-execute cycle.
3. **"Explain it Like I'm 5":** Groups explain their assigned component in simple terms to reinforce understanding.

5. Differentiation Summary

Level	Strategy
Support	Provide pre-labelled diagrams and sentence stems for the FFF paragraph. Supply the initial "Feature" so students focus on evidence and interaction.
Challenge	Research alternative architectures or sketch a diagram showing how a co-processor (GPU) alters data flow.

6. Extension Activities

- **CPU Simulator:** Use *Little Man Computer* to visualise the fetch-decode-execute cycle.
- **Real-World Analysis:** Compare CPU specs (cores, cache, clock speed) across devices.
- **Design Project:** Design a hypothetical CPU tailored to a specific application (e.g., gaming, AI).

7. Resources

- Presentation (Slide Deck)
 - InThinking eBook (A 1.1.1)
 - Worksheet (Blank CPU Diagram)
 - Quizlet (Vocabulary Practice)
 - Little Man Computer (Online Simulator)
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Author's Reflection

"I used to teach the physical components and the machine instruction cycle as separate entities. Bringing them together links the physical components directly to their functions in the cycle. By focusing on disciplinary literacy, I have sacrificed some initial role-play activities. However, ensuring students accurately use Tier 3 vocabulary pays much higher dividends in final written assessments."